



ĐẠI HỌC ĐÀ NẴNG
TRƯỜNG ĐẠI HỌC CÔNG NGHỆ THÔNG TIN VÀ TRUYỀN THÔNG VIỆT - HÀN
Vietnam - Korea University of Information and Communication Technology

BUILDING A CROSS-PLATFORM INTELLIGENT TOURISM SYSTEM USING AI

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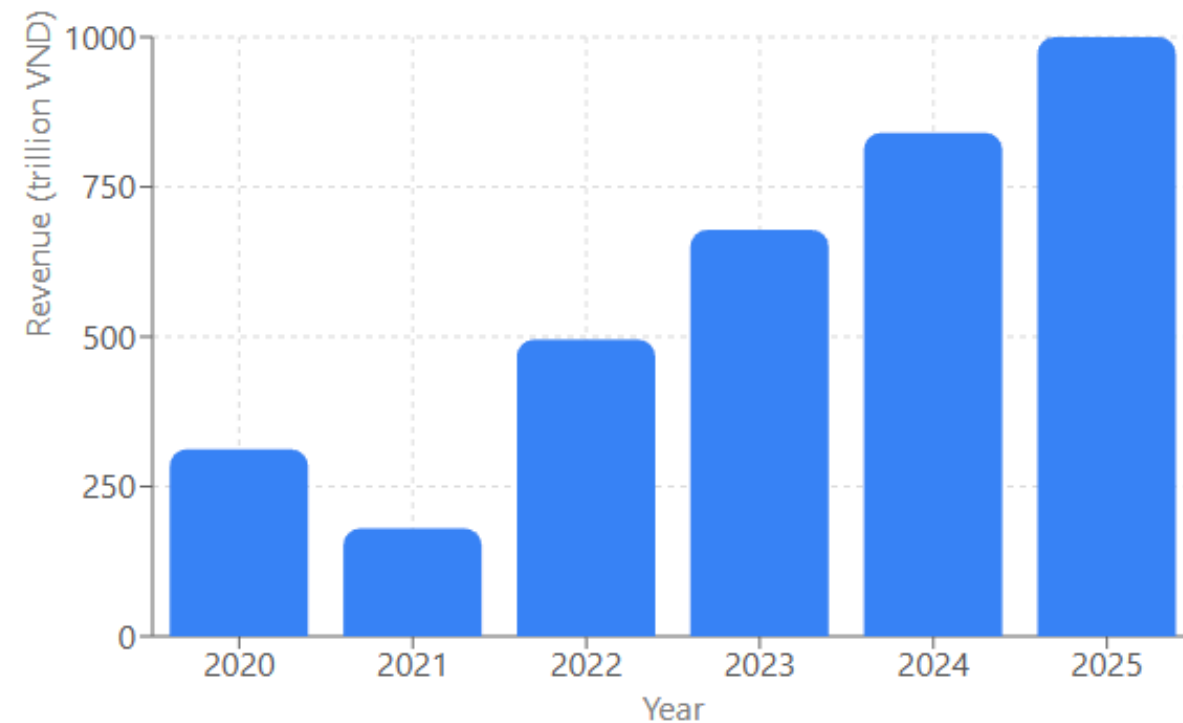
PRESENTATION OUTLINE

- 1. Context & Problem Statement**
- 2. Technologies Used**
- 3. System Analysis and Design**
- 4. AI Technologies**
- 5. Conclusion & Demo**



1. Context & Problem Statement

TOURISM REVENUE OVER THE YEARS



- Tourism strongly recovered after COVID
- Increasing demand for travel support apps

● INCONVENIENCES WHEN PLANNING TRIPS:

✗ SCATTERED INFORMATION

- Hotels: Booking.com, Agoda
 - Tours: Klook, GetYourGuide
 - Restaurants: Google Maps
- Need to use multiple separate apps

✗ GENERIC RECOMMENDATIONS

- Apps often suggest the same for everyone
- Not based on personal preferences



1. Context & Problem Statement

PROJECT OBJECTIVES - TRIPFINITY:

BUILD AN INTEGRATED APPLICATION

Combine 4 service types in 1 app:

 Hotel |  Restaurant |  Attraction |  Tour

APPLY AI RECOMMENDATION

- Recommend services based on user behavior
- Prioritize by location, price, service features

BUILD SUPPORT CHATBOT

- Search services using natural language
- Respond based on system data



2. Technologies Used

FRONTEND

- Flutter 3.x - Mobile App (Tourist)
- React + TypeScript - Web Portal (Supplier)

BACKEND

- Spring Boot 3.5 + Java 17 - RESTful API
- Python + Flask - AI Server

DATABASE

- MySQL 8.0 - 3NF normalized

AI/ML

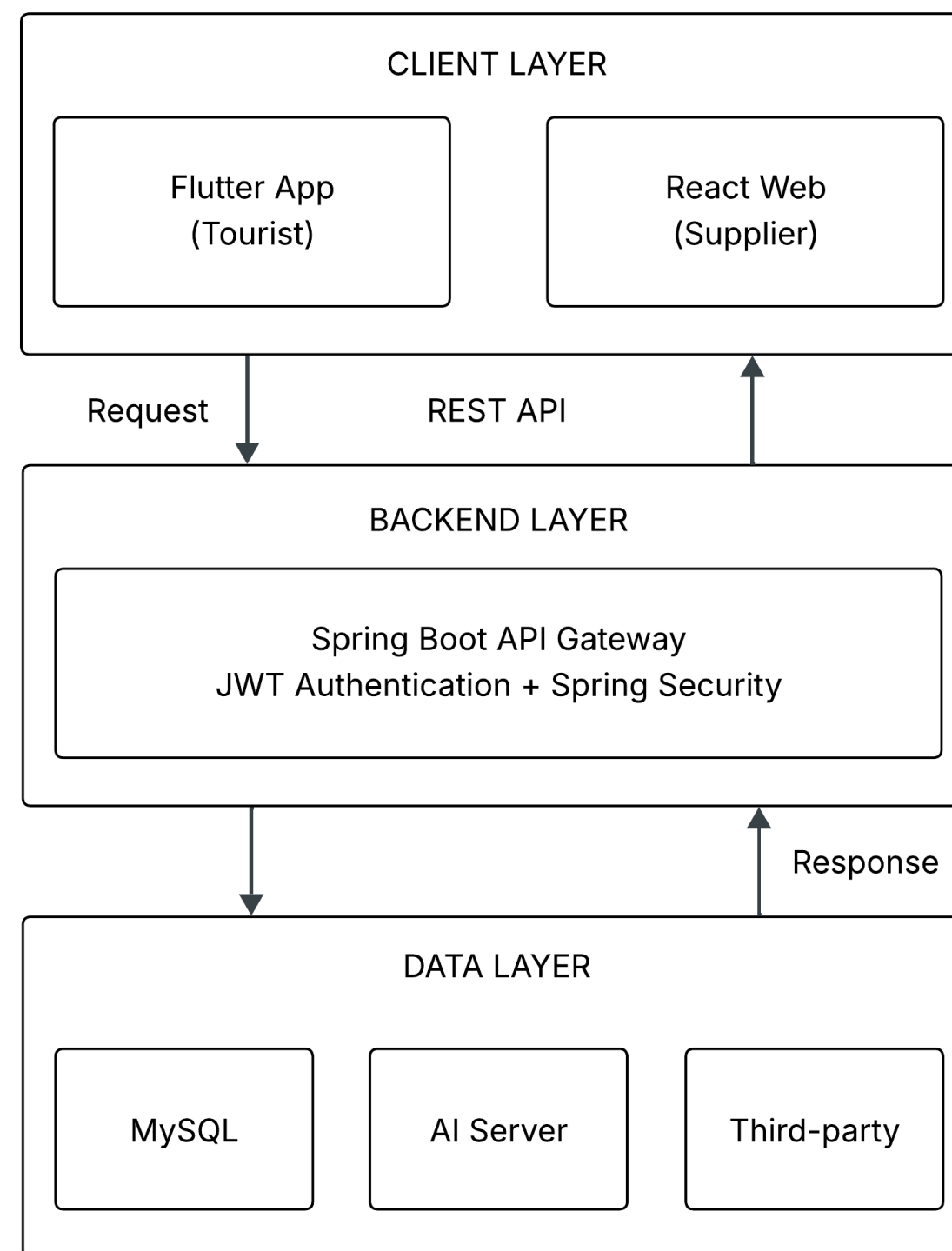
- Two-Tower Model (Recommendation) - SELF-BUILT
- LangChain + Llama-3 70B via Groq (Chatbot RAG)

THIRD-PARTY INTEGRATIONS

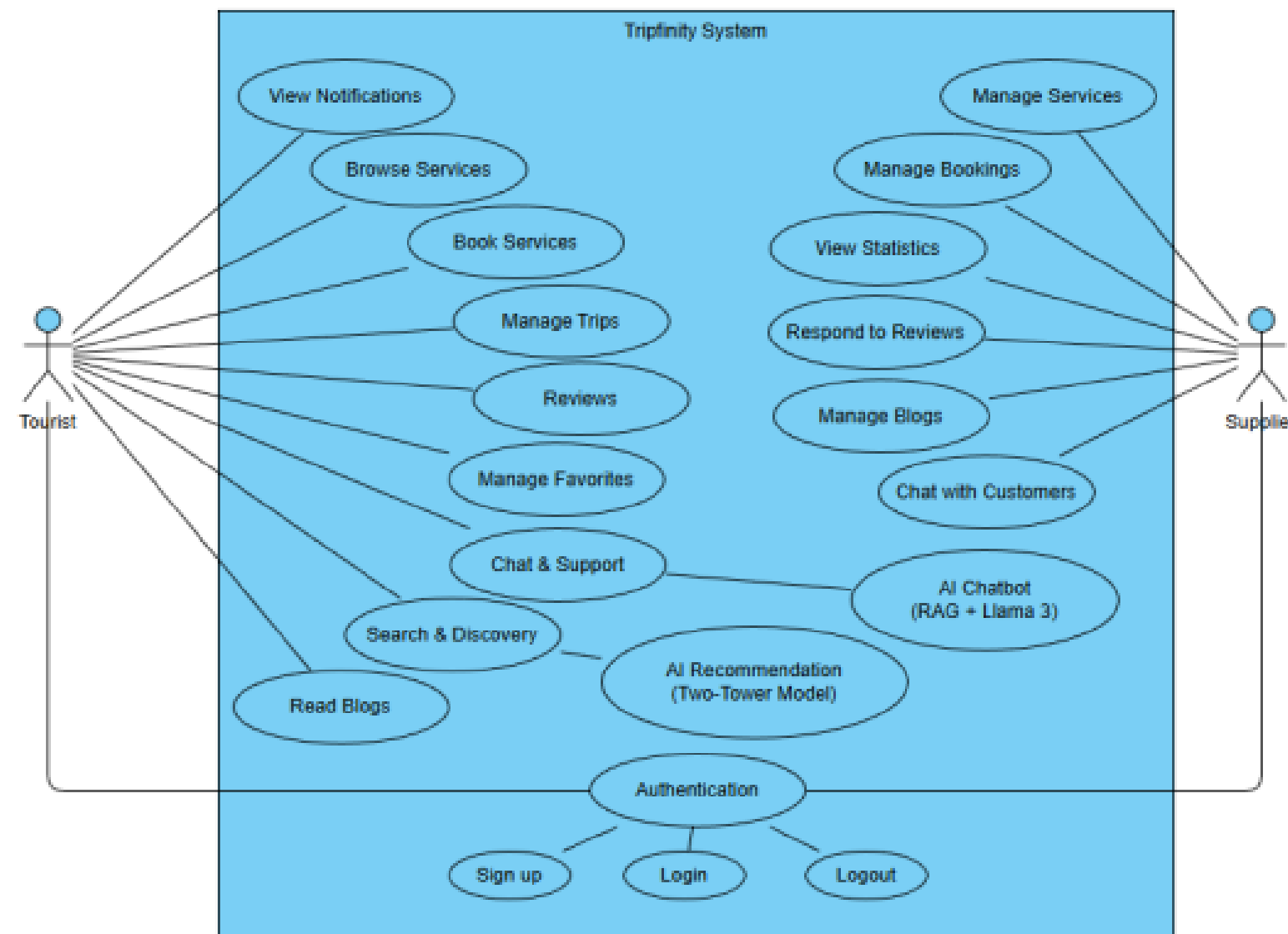
- Firebase | ZaloPay | Google Maps | Cloudinary

3. System Analysis and Design

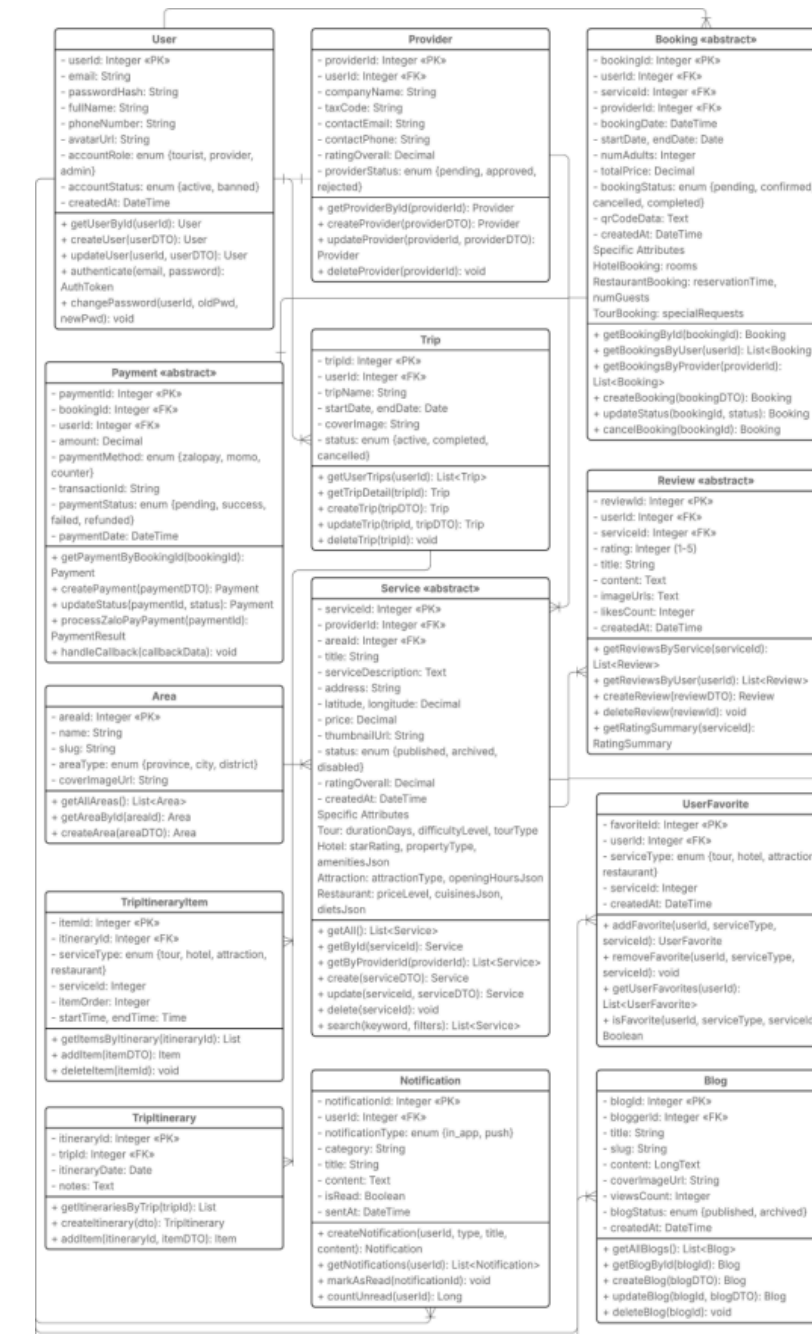
System Architecture



3. System Analysis and Design

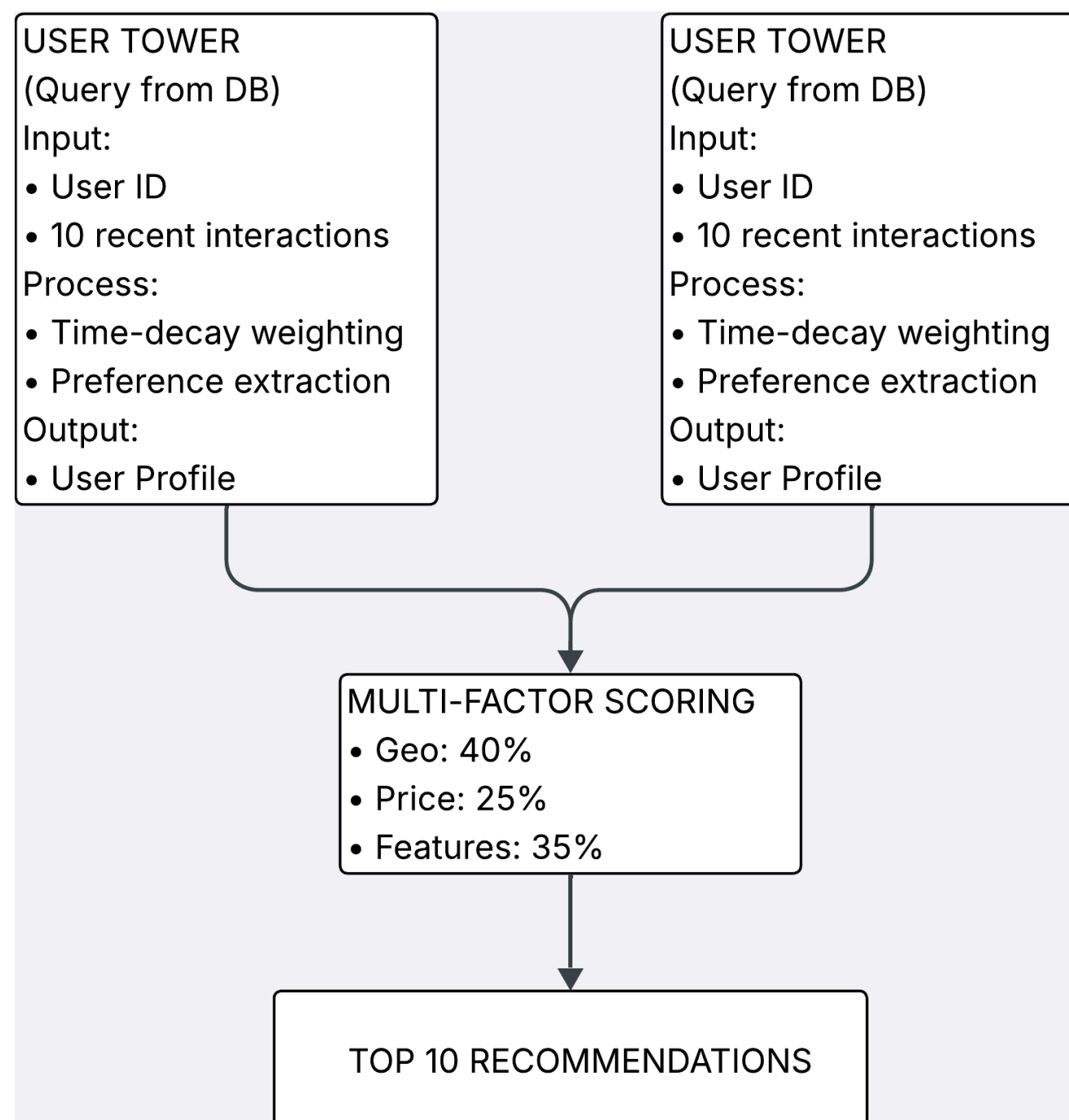


Use case Diagram



Class Diagram

4. AI Technologies



TWO-TOWER MODEL (Real-time Inference - Self-built)

HOTEL

- **amenities_json:** WiFi, Pool, Spa, Gym, Parking...
- **property_type:** Hotel, Resort, Villa, Homestay
- **star_rating:** 1-5 stars (hotel class)

RESTAURANT

- **cuisines_json:** Vietnamese, Japanese, Italian...
- **diets_json:** Vegetarian, Vegan, Halal, Gluten-free
- **categories_json:** Fine Dining, Street Food, Cafe

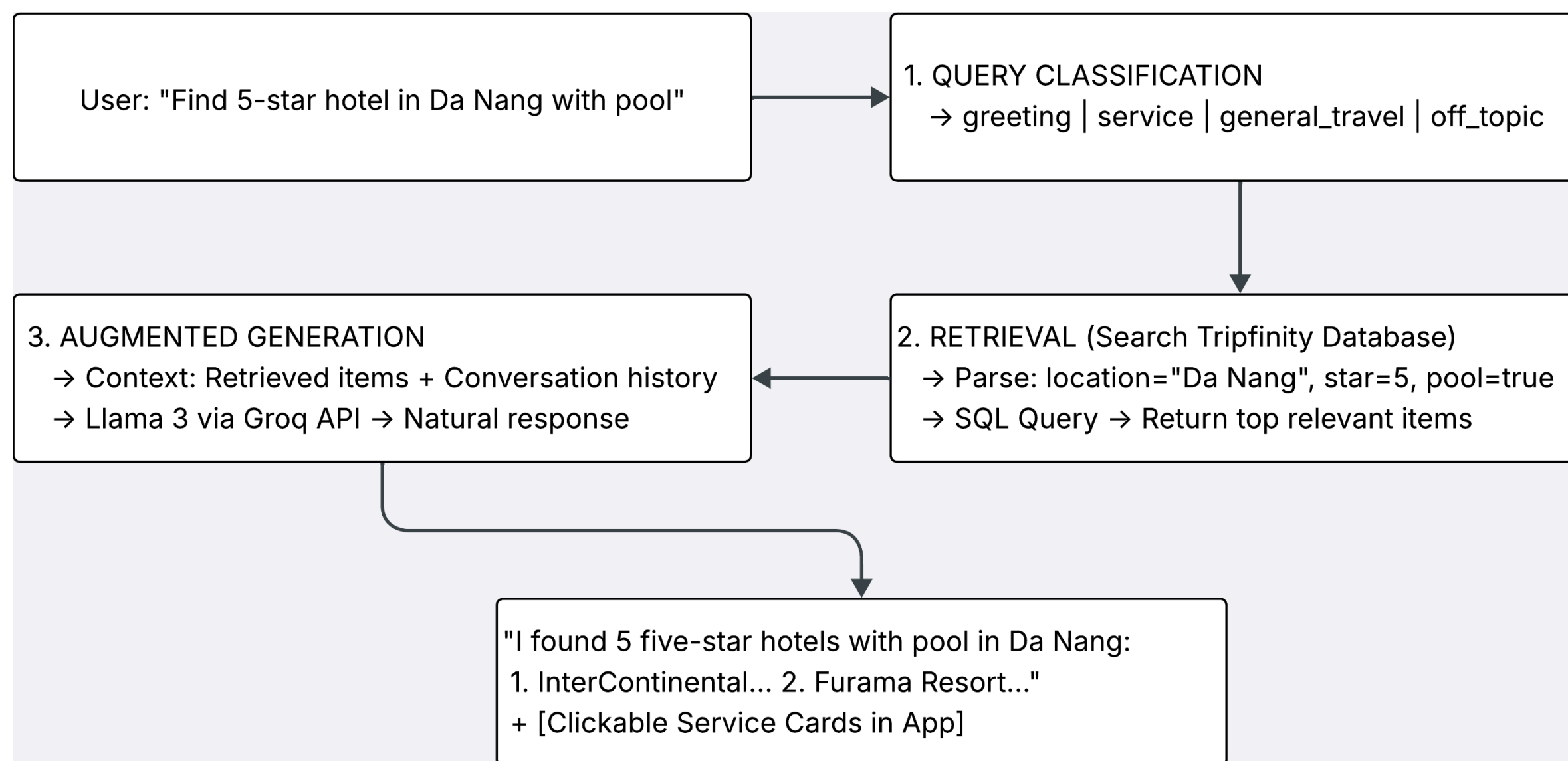
ATTRACTION

- **attraction_type:** Museum, Temple, Park, Beach...
- **suitable_for_json:** Family, Couples, Solo, Kids
- **categories_json:** Cultural, Nature, Entertainment

TOUR

- **tour_type:** Group, Private, Custom
- **difficulty_level:** Easy, Moderate, Hard
- **categories_json:** Adventure, Cultural, Nature

4. AI Technologies



TRIPBOT - RAG + LLAMA 3 (70B Parameters)



5. Conclusion & Demo

✓ ACHIEVEMENTS:

📱 MOBILE APP (Flutter):

- 51 complete screens
- Cross-platform: Android, iOS, Web

💻 WEB SUPPLIER (React + TypeScript):

- Full CRUD for 4 service types
- Statistics dashboard

🤖 AI MODULES:

- Two-Tower: Real-time inference
- Chatbot: Multilingual
- Cold Start Problem solved

💻 BACKEND:

- 36+ REST API endpoints
- 39 database tables (3NF)

💳 INTEGRATIONS:

- ZaloPay Payment ✓
- Firebase Auth + FCM ✓
- Google Maps ✓
- Cloudinary ✓



5. Conclusion & Demo

LIMITATIONS:

- Not yet deployed to production
- No formal AI benchmark
- Not integrated with real booking data sources

FUTURE DEVELOPMENT:

- Deploy to cloud (AWS/GCP) with CI/CD
- A/B Testing to optimize AI weights
- Integrate VNPay, Momo payment
- Multi-modal search (voice, image)



PROJECT DEMO



THANK YOU

Q&A SESSION

